K642 — Down-Row Disposition Settled: Running-Immune m_s/m_d Kills the Scale Defense; It's a GUT-FREE Structural MISS, Not a Five-Absence Failure

Auditor: Keeper | Date: 2026-07-02 Mid-Year (long pull) | Verdict: down-row = MISS / STRUCTURAL (not "conditional-pending-scale"); NOT a Five-Absence violation | Count: 8 total unchanged; honest tier now 3 firm + 3 structural-MISS + α partial + muon

Settles the K641 CONDITIONAL flag using two independent long-pull findings (Elie 4540, Grace steelman) that pull in opposite directions and together give a cleaner, more defensible disposition than K641's two-horn framing.

1. Elie 4540 — the running-immune nail (CONFIRMED)

The down-row banks + the muon bank force a same-sector ratio: - m_d/m_e = 3, m_s/m_μ = 1/3, m_μ/m_e = $(24/\pi^2)^6$ = 206.76 (verified 0.004%, RG-invariant) - ⍰ m_s/m_d = (1/3 · 206.76)/3 = 22.97 - Observed m_s/m_d ≈ 20.0, which is RG-INVARIANT (same-sector down-quark ratio, K631/F444) → no scale/running choice can move it. - Tension = 14.9%, running-immune.

Independently verified (Keeper). CORRECTION (Elie re-pin, 2026-07-02): I over-elevated this as "the nail." PDG 2024 gives m_s/m_d = 20.0 ± 2.4 (~12% error), so 14.9% deviation is only ~1.24σ — corroboration, not a decisive kill. dev% ≠ σ-significance, and I quoted 20 as if exact. The running-immune cross-check *supports* the disposition but is not what settles it. What settles it: the direct value-misses — the down-row ratios miss authoritative PDG at 6.5σ / 6.8σ / 80σ (Elie 4543). Those are decisive; the m_s/m_d cross-check is a ~1σ echo.

2. Grace steelman — Horn A CLEARED (it is NOT a Five-Absence violation)

Grace steelmanned her own flag and partially cleared it: BST forces a substrate/Planck scale (≠ M_GUT, no GUT machinery); because QCD running saturates, that substrate scale gives ~the same low-scale result M_GUT would. So the "GUT-scale adjacency" is a saturation artifact, not a smuggled GUT. She withdrew the "GUT-adjacency kills it" framing. Concur: the down-row mechanism (color-root parity) is genuinely GUT-free; this is not a Five-Absence failure. Good discipline — the bar fired against her own flag.

3. Net disposition (cleaner than K641)

The two findings combine into the honest, referee-defensible story:

> The down-row is a genuinely GUT-free substrate texture (Grace's mechanism, Horn A cleared) that is ~15% internally inconsistent with data on a running-immune ratio (Elie's m_s/m_d, Horn B floor). It is not a firm bank and not a Five-Absence violation — it is a structural MISS.

- The scorecard already scores the individual ratios MISS (m_d/m_e 67%, m_s/m_μ 62%, m_b/m_τ 57% vs observed) — the tool's verdict and the internal cross-check agree.
- Re-tier: down-row FIRM → STRUCTURAL-MISS (supersedes K641's CONDITIONAL — scale can't clear it, so "conditional pending scale" was too generous). In the 26-derivation loop it is a MISS: either a new mechanism forces the *observed* {9.1, 0.88, 2.35} (not the GJ {3,1/3,1}), or it terminates as a structural texture with the 15% stated honestly.
- Count: 8 total unchanged (they remain derivations of *something*, just not firm banks). Honest tier: 3 firm (θ_QCD, m_t, θ_{13}) + 3 structural-MISS (down-row) + α partial + muon principle-gated. Staged Casey-ratification-pending.

4. New failure mode for the tool — internal-consistency cross-checks

The m_s/m_d kill is a failure mode the per-parameter match does NOT catch: banks that individually pass (or are claimed) can combine to force a THIRD observable that fails. m_s/m_d is derivable from three banks and lands 15% off. Proposed Pass-1 tool enhancement: add cross-consistency checks to `bst_almanac.py` — every ratio derivable from ≥2 banks gets scored too. This would have caught the down-row from the muon+down-row banks alone, without needing the GJ insight.

5. Lyra’s Pauli / color-Fock finding (separate thread) — brief audit

Pauli exclusion = exterior-algebra nilpotency (v∧v=0); SO(7) Dirac spinor $\Lambda(\mathbb{C}^3)$ *= 2^{N_c} = 8 = the color Fock space; confinement = only* Λ^3 *(singlet volume) manifests. Honest tier confirmed: the exterior-algebra facts + “SO(7) spinor* = $\Lambda(\mathbb{C}^3)$” are solid standard math; “the $\mathbb{C}^3$ IS the color frame” is a conjecture (handed to Grace: is color’s $\mathbb{C}^3$ the maximal isotropic subspace of so(7)?). No new number → no bank, count stays 8. It’s a conceptual unification (understanding), not a derivation — it does not enter the 26-loop. On “new for nuclear physics”: that’s Lyra’s literature check — exterior-algebra-of-color and geometric-confinement reformulations have precedents; don’t over-claim novelty until she checks. Her thread per Casey’s direction.

— Keeper K642, Mid-Year 2026-07-02. Down-row settled: GUT-free (Grace clears Five-Absence) but running-immune 15% inconsistent (Elie m_s/m_d) → STRUCTURAL-MISS, not firm, not a Five-Absence failure. Count 8; honest tier 3 firm + 3 structural + α + muon. Cross-consistency check proposed for the tool. Lyra’s Pauli finding: conceptual, no bank.